Workshop Practice Lab:

Workshop Practice Lab is a practical course that introduces students to the basic techniques and tools used in various engineering workshops. The course is designed to provide hands-on experience in the use of workshop tools and equipment, and to develop skills in the fabrication of simple engineering components. The following are some of the key topics covered in the Workshop Practice Lab:

1. Safety: Students are taught the importance of safety in the workshop, including the use of personal protective equipment, safe handling of tools and equipment, and the proper storage of materials.
2. Hand Tools: Students learn to identify and use common hand tools such as hammers, screwdrivers, pliers, and wrenches.
3. Measuring Instruments: Students learn to use measuring instruments such as vernier calipers, micrometers, and dial gauges to take accurate measurements.
4. Machine Tools: Students are introduced to basic machine tools such as lathes, milling machines, and drilling machines, and learn to operate them safely.
5. Welding: Students learn the basics of welding, including the use of welding equipment, the selection of welding rods, and the proper techniques for welding different types of metals.
6. Sheet Metal Work: Students learn to cut, bend, and shape sheet metal using hand tools and machines.
7. Fitting: Students learn to fit and assemble mechanical components using techniques such as filing, sawing, and drilling.
8. Carpentry: Students learn the basics of carpentry, including the use of hand and power tools, and the construction of simple wooden structures.

Overall, the Workshop Practice Lab provides students with a solid foundation in the practical skills required in engineering workshops, and prepares them for more advanced courses in their chosen field of study.

S. No. Mechanical Workshop Duration

1. Introduction to Mechanical Workshop: A mechanical workshop is a place where various mechanical operations are performed using different tools and machines. The duration of a mechanical workshop can vary depending on the scope and objectives of the workshop.
2. Types of Mechanical Workshops: There are different types of mechanical workshops, such as machining workshops, welding workshops, and maintenance workshops. The duration of these workshops can vary depending on the type of workshop and the skills being taught.

3. Factors Affecting Workshop Duration: The duration of a mechanical workshop can be affected by several factors, including the number of participants, the complexity of the tasks being performed, and the level of expertise of the participants.

4. Planning a Mechanical Workshop: When planning a mechanical workshop, it is important to consider the duration of the workshop to ensure that there is enough time to cover all the necessary topics and activities. The duration should be balanced with the objectives of the workshop to ensure that the participants get the most out of the experience.

5. Conclusion: The duration of a mechanical workshop can vary depending on several factors. It is important to carefully plan the duration of the workshop to ensure that the objectives of the workshop are met and the participants have a positive and productive experience.

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

A mechanical workshop is a place where various materials, tools, and machines are used to manufacture, repair, or maintain mechanical systems. In this section, we will introduce some of the common materials, tools, and machines found in a mechanical workshop.

Materials

- **Metals**: Metals such as steel, aluminum, and copper are commonly used in mechanical workshops due to their strength, durability, and versatility.
- **Plastics**: Plastics are also commonly used in mechanical workshops due to their lightweight, low cost, and ease of fabrication.
- **Wood**: Wood is used in mechanical workshops for its natural beauty, strength, and workability.

Tools

- **Hand tools**: Hand tools such as hammers, screwdrivers, and pliers are commonly used in mechanical workshops for tasks such as fastening, cutting, and shaping.
- **Power tools**: Power tools such as drills, saws, and grinders are used in mechanical workshops to make tasks easier and more efficient.
- **Measuring tools**: Measuring tools such as rulers, calipers, and micrometers are used in mechanical workshops to ensure accuracy and precision in the manufacturing process.

Machines

- **Lathes**: Lathes are used in mechanical workshops to shape and form materials such as metal, wood, and plastic.

- **Milling machines**: Milling machines are used in mechanical workshops to cut and shape materials using rotary cutters.
- **Drilling machines**: Drilling machines are used in mechanical workshops to create holes in materials using drill bits.

In summary, a mechanical workshop is a place where various materials, tools, and machines are used to manufacture, repair, or maintain mechanical systems. Understanding the common materials, tools, and machines found in a mechanical workshop is essential for anyone working in the field of mechanics.

Layout, Safety Measures and Different Engineering Materials Used in Workshop

The layout of a workshop refers to the arrangement of equipment, machinery, and workstations in a manner that optimizes the flow of work and enhances safety. A well-designed workshop layout can improve productivity, reduce the risk of accidents, and increase the overall efficiency of the workshop.

Safety measures in a workshop are essential to protect workers from potential hazards and to prevent accidents. Some common safety measures include the use of personal protective equipment (PPE), proper ventilation, and the implementation of safety protocols and procedures.

Different engineering materials are used in a workshop, including mild steel, medium carbon steel, high carbon steel, high speed steel, and cast iron. Each of these materials has unique properties that make them suitable for specific applications.

- **Mild steel** is a low carbon steel that is commonly used in construction and manufacturing due to its strength, ductility, and affordability.
- **Medium carbon steel** has a higher carbon content than mild steel, which increases its strength and hardness. It is commonly used in the production of machine parts and tools.
- **High carbon steel** has an even higher carbon content than medium carbon steel, which further increases its strength and hardness. It is commonly used in the production of cutting tools and springs.
- **High speed steel** is a type of tool steel that is used in the production of cutting tools due to its ability to maintain its hardness at high temperatures.
- **Cast iron** is an alloy of iron, carbon, and silicon that is known for its durability and resistance to wear. It is commonly used in the production of engine blocks and other heavy machinery components.

It is important to select the appropriate engineering material for a specific application in order to ensure the safety and efficiency of the workshop.

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

Fitting, sheet metal, and welding are essential processes in the manufacturing and construction industries. These processes require the use of various tools, equipment, devices, and machines to achieve the desired results. Below is a list of some of the most commonly used tools, equipment, devices, and machines in these sections:

1. **Fitting tools:** These include tools such as hammers, chisels, files, hacksaws, and wrenches. These tools are used to shape, cut, and join metal parts.
2. **Sheet metal tools:** These include tools such as shears, punches, and bending brakes. These tools are used to cut, shape, and bend sheet metal.
3. **Welding equipment:** This includes equipment such as welding machines, welding torches, and welding electrodes. These are used to join metal parts by melting and fusing them together.
4. **Measuring and marking tools:** These include tools such as rulers, calipers, and scribers. These tools are used to measure and mark metal parts to ensure accuracy and precision.
5. **Safety equipment:** This includes equipment such as gloves, goggles, and respirators. These are used to protect workers from the hazards associated with fitting, sheet metal, and welding processes.

It is important to note that the specific tools, equipment, devices, and machines used in fitting, sheet metal, and welding sections may vary depending on the specific task and the materials being used. It is essential to select the appropriate tools and equipment for the job to ensure safety and efficiency.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

1. **Vernier Calliper**: The least count of a vernier calliper is determined by dividing the smallest division on the main scale by the total number of divisions on the vernier scale. For example, if the smallest division on the main scale is 1mm and the total number of divisions on the vernier scale is 10, then the least count of the vernier calliper is $1\text{mm}/10 = 0.1\text{mm}$.
2. **Vernier Height Gauge**: The least count of a vernier height gauge is determined in the same way as that of a vernier calliper. The smallest division on the main scale is divided by the total number of divisions on the vernier scale.
3. **Micrometer (Screw Gauge)**: The least count of a micrometer is determined by dividing the pitch of the screw by the total number of divisions on the thimble. For example, if the pitch of the screw is 0.5mm and the total number of divisions on the thimble is 50, then the least count of the micrometer is $0.5\text{mm}/50 = 0.01\text{mm}$.

To take different readings over given metallic pieces using these instruments, the following steps can be followed:

1. Place the metallic piece between the jaws of the vernier calliper or the anvil and spindle of the micrometer.
2. Close the jaws or the anvil and spindle until they touch the metallic piece.
3. Read the measurement on the main scale and add it to the measurement on the vernier scale or thimble to get the final reading.
4. Repeat the process for different metallic pieces and record the readings.

Machine Shop (3 Hours)

A machine shop is a facility where machining is performed. Machining is the process of removing material from a workpiece to create a desired shape or surface. This is typically done using machine tools such as lathes, milling machines, and drill presses. Here are some key points to consider when studying machine shops:

1. **Machine tools**: Machine shops typically contain a variety of machine tools, each designed for a specific type of machining operation. Common machine tools include lathes, milling machines, drill presses, and grinders.

2. **Workholding**: Workholding refers to the methods used to secure a workpiece in place while it is being machined. Common workholding methods include vises, clamps, and chucks.

3. **Cutting tools**: Cutting tools are used to remove material from the workpiece. These tools come in a variety of shapes and sizes, and are typically made from materials such as high-speed steel or carbide.

4. **Coolants and lubricants**: Coolants and lubricants are used to reduce heat and friction during the machining process. This can help to extend the life of the cutting tools and improve the quality of the finished workpiece.

5. **Safety**: Safety is an important consideration in any machine shop. Operators must be trained to use the machine tools safely, and appropriate personal protective equipment (such as safety glasses and hearing protection) should be worn at all times.

6. **Quality control**: Quality control is the process of ensuring that the finished workpiece meets the desired specifications. This can involve the use of measuring tools such as micrometers and calipers, as well as visual inspection and testing.

7. **Maintenance**: Regular maintenance is important to keep the machine tools in a machine shop running smoothly. This can involve tasks such as cleaning, lubrication, and replacement of worn or damaged parts.

In summary, a machine shop is a facility where machining operations are performed using a variety of machine tools. Key considerations when studying machine shops include the types of machine tools used, workholding methods, cutting tools, coolants and lubricants, safety, quality control, and maintenance.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

A lathe is a machine tool that rotates a workpiece about an axis of rotation to perform various operations such as cutting, sanding, knurling, drilling, deformation, facing, and turning. The lathe is one of the oldest and most important machine tools.

The main parts of a lathe machine are the headstock, tailstock, bed, carriage, and lead screw. The headstock contains the spindle and the drive mechanism for rotating the workpiece. The tailstock supports the other end of the workpiece. The bed is the base of the machine and provides support for the other components. The carriage moves along the bed and supports the cutting tool. The lead screw is used to move the carriage along the bed.

Accessories for a lathe machine include chucks, faceplates, centers, and tool holders. Chucks are used to hold the workpiece in place. Faceplates are used to hold irregularly shaped workpieces. Centers are used to support the workpiece between the headstock and tailstock. Tool holders are used to hold the cutting tool.

Facing is the operation of machining the ends of a workpiece to produce a flat surface. Plane turning is the operation of machining a cylindrical surface. Step turning is the operation of machining a stepped cylindrical surface. Taper turning is the operation of machining a conical surface. Threading is the operation of cutting a screw thread. Knurling is the operation of producing a roughened surface on a workpiece. Parting is the operation of cutting off a piece of the workpiece.

These are the basic operations that can be performed on a lathe machine. Each operation requires a different set of tools and techniques. It is important to understand the construction and accessories of a lathe machine in order to perform these operations effectively.

Fitting Shop

A fitting shop is a workshop where the process of fitting takes place. Fitting is the process of assembling, shaping, and finishing parts to create a final product. This process is typically carried out by skilled workers using a variety of hand and power tools.

Here are some key points to know about fitting shops:

1. Fitting shops are typically found in manufacturing and engineering industries, where they play a crucial role in the production process.
2. The work carried out in a fitting shop can vary greatly depending on the industry and the specific product being produced. Some common tasks include cutting, drilling, grinding, and shaping metal or other materials.
3. Fitting shops are typically equipped with a range of tools and machinery, including lathes, milling machines, drill presses, and grinders.
4. Safety is a top priority in fitting shops, as the work can be dangerous if proper precautions are not taken. Workers are typically required to wear protective equipment such as safety glasses, gloves, and earplugs.
5. Fitting shops may also employ quality control measures to ensure that the final product meets the required specifications and standards.

1. Practice marking operations

- Marking operations are an essential part of many manufacturing processes.
- They are used to identify, label, and track parts and products.
- Marking can be done using various methods, including stamping, engraving, etching, and printing.
- The choice of marking method depends on factors such as the material being marked, the desired mark quality, and the production volume.
- It is important to practice marking operations to ensure that the marks are accurate, legible, and consistent.
- Practice can involve testing different marking methods and parameters to determine the best approach for a given application.
- Regular practice can also help to identify and address any issues with the marking equipment or process.
- By practicing marking operations, manufacturers can improve the quality and efficiency of their marking processes, leading to better product identification and tracking.

2. Preparation of U or V -Shape Male Female Work piece

The preparation of U or V -Shape Male Female Work piece involves several steps, including Filing, Sawing, Drilling, and Grinding. Below is a detailed explanation of each step:

1. **Filing:** Filing is the process of smoothing, shaping, or removing material from a workpiece using a file. A file is a tool with a rough surface used to remove small amounts of material from a workpiece. Filing is typically used to remove burrs or rough edges from a workpiece.

2. **Sawing:** Sawing is the process of cutting a workpiece using a saw. A saw is a tool with a toothed blade used to cut through materials such as wood, metal, or plastic. Sawing is typically used to cut a workpiece to the desired size or shape.

3. **Drilling:** Drilling is the process of creating a hole in a workpiece using a drill. A drill is a tool with a rotating cutting edge used to create holes in materials such as wood, metal, or plastic. Drilling is typically used to create holes for fasteners such as screws or bolts.

4. **Grinding:** Grinding is the process of removing material from a workpiece using an abrasive wheel. An abrasive wheel is a tool with a rough surface used to remove material from a workpiece. Grinding is typically used to smooth or shape a workpiece, or to remove material from a workpiece to achieve a desired size or shape.

These steps are used to prepare a U or V -Shape Male Female Work piece to the desired specifications. Each step is important and must be performed with care to ensure the final workpiece meets the desired standards.

4 Carpentry Shop 3 Hrs

1. A carpentry shop is a workshop where carpenters work with wood to create various objects such as furniture, cabinets, and other wooden structures.
2. Carpentry is a skilled trade that involves cutting, shaping, and joining wood to construct buildings, furniture, and other structures.
3. In a carpentry shop, carpenters use various tools such as saws, hammers, chisels, and planes to shape and join wood.
4. A carpentry shop may also have machinery such as table saws, band saws, and jointers to assist in the woodworking process.
5. The time spent in a carpentry shop can vary depending on the project being worked on. In this case, 3 hours is the time allotted for work in the carpentry shop.
6. During this time, carpenters may work on various tasks such as measuring and cutting wood, assembling pieces, and sanding and finishing the final product.
7. Carpentry requires precision and attention to detail, and the time spent in the shop allows carpenters to carefully craft their work to meet the desired specifications.
8. The end result of the time spent in the carpentry shop is a finished product that is both functional and aesthetically pleasing.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

Carpentry is a skilled trade and a craft in which the primary work performed is the cutting, shaping and installation of building materials during the construction of buildings, ships, timber bridges, concrete formwork, etc. Carpenters traditionally worked with natural wood and did the rougher work such as framing, but today many other materials are also used and sometimes the finer trades of cabinetmaking and furniture building are considered carpentry.

To perform carpentry, various tools and equipment are required. Some of the common carpentry tools and equipment include:

1. **Saw**: A saw is a tool consisting of a tough blade, wire, or chain with a hard toothed edge. It is used to cut through material, very often wood though sometimes metal or stone.
2. **Hammer**: A hammer is a tool consisting of a weighted "head" fixed to a long handle that is swung to deliver an impact to a small area of an object.
3. **Chisel**: A chisel is a tool with a characteristically shaped cutting edge of blade on its end, for carving or cutting a hard material such as wood, stone, or metal by hand, struck with a mallet, or mechanical power.
4. **Plane**: A hand plane is a tool for shaping wood using muscle power to force the cutting blade over the wood surface.
5. **Measuring tape**: A tape measure or measuring tape is a flexible ruler used to measure size or distance.

In carpentry, various joints are used to connect two or more pieces of wood. Some of the common joints used in carpentry include:

1. **Cross Half Lap Joint**: A cross half lap joint is made by removing material from each of two pieces of wood to half their thickness where they will cross each other. The two pieces are then fitted together to form a strong joint.
2. **Half Lap Dovetail Joint**: A half lap dovetail joint is similar to a cross half lap joint, but with the addition of a dovetail shape to provide additional strength and resistance to pulling apart.
3. **Mortise and Tenon Joint**: A mortise and tenon joint is a type of joint where the end of one piece of wood (the tenon) is inserted into a hole (the mortise) in another piece of wood. The tenon is then secured in place using glue, pins, or wedges.

To make a cross half lap joint, follow these steps:

1. Mark the position of the joint on both pieces of wood.
2. Use a saw to cut away half the thickness of each piece of wood at the joint location.
3. Fit the two pieces together to form the joint.

To make a half lap dovetail joint, follow these steps:

1. Mark the position of the joint on both pieces of wood.
2. Use a saw and chisel to cut away half the thickness of each piece of wood at the joint location, creating a dovetail shape.
3. Fit the two pieces together to form the joint.

To make a mortise and tenon joint, follow these steps:

1. Mark the position of the joint on both pieces of wood.

2. Use a saw and chisel to cut a mortise (hole) in one piece of wood and a tenon (protruding end) on the other piece of wood.
3. Fit the tenon into the mortise and secure in place using glue, pins, or wedges.

These are some of the basic techniques used in carpentry to create strong and secure joints between pieces of wood. With practice and experience, a carpenter can learn to create a wide variety of joints to suit different applications.

5 Welding Shop 6 Hrs

Welding is a fabrication process that joins two or more parts by heating the surfaces to their melting points and applying pressure to fuse them together. Welding is used in a variety of industries, including construction, manufacturing, and repair.

1. **Types of Welding:** There are several types of welding, including arc welding, gas welding, and resistance welding. Each type of welding uses different techniques and equipment.

2. **Welding Equipment:** Welding equipment includes welding machines, electrodes, filler materials, and protective gear. Welding machines generate heat and electricity to melt the metal, while electrodes and filler materials are used to join the parts together.

3. **Welding Safety:** Welding can be dangerous if proper safety precautions are not taken. Welders must wear protective gear, including gloves, masks, and goggles, to protect themselves from heat, sparks, and fumes.

4. **Welding Techniques:** Welding techniques vary depending on the type of welding being performed. Some common welding techniques include stick welding, MIG welding, and TIG welding.

5. **Welding Applications:** Welding is used in a variety of applications, including construction, manufacturing, and repair. Welding is used to join metal parts in buildings, bridges, vehicles, and machinery.

Introduction to BI standards and reading of welding drawings

BI standards refer to the standards set by the British Institute of Standards for welding and fabrication. These standards provide guidelines for the design, fabrication, and inspection of welded joints.

Reading welding drawings is an essential skill for anyone involved in the welding and fabrication industry. Welding drawings provide detailed information about the dimensions, materials, and welding processes to be used in the fabrication of a component.

In order to read welding drawings, one must be familiar with the standard symbols and conventions used to represent welding information. These symbols provide information about the type of weld, the size of the weld, and the location of the weld.

Practice of Making following operations:

1. **Butt Joint:** A butt joint is a type of joint where two pieces of material are joined end to end. This type of joint is commonly used in welding and can be created using a variety of welding processes.

2. **Lap Joint:** A lap joint is a type of joint where two pieces of material are overlapped and joined together. This type of joint is commonly used in welding and can be created using a variety of welding processes.

3. **TIG Welding:** TIG welding, also known as Gas Tungsten Arc Welding (GTAW), is a welding process that uses a non-consumable tungsten electrode to produce the weld. This process is commonly used for welding thin materials and for creating precise welds.

4. **MIG Welding:** MIG welding, also known as Gas Metal Arc Welding (GMAW), is a welding process that uses a consumable wire electrode to produce the weld. This process is commonly used for welding thicker materials and for creating strong welds quickly.

6 Moulding and Casting Shop 6 Hrs

Moulding and casting are manufacturing processes used to produce complex shapes. These processes involve pouring molten metal into a mould, which contains a hollow cavity of the desired shape, and then allowing the metal to cool and solidify. The solidified part is then removed from the mould to complete the process.

1. Moulding is the process of manufacturing by shaping liquid or pliable raw material using a rigid frame called a mould.
2. Casting is a manufacturing process in which a liquid material is usually poured into a mould, which contains a hollow cavity of the desired shape, and then allowed to solidify.
3. The most common metals processed are aluminium and cast iron. However, other metals, such as bronze, brass, steel, magnesium, and zinc, are also used to produce castings.
4. There are two main types of moulding: sand casting and die casting.
5. Sand casting involves creating a mould from a sand mixture and then pouring molten metal into the mould.
6. Die casting involves forcing molten metal under high pressure into a mould cavity.
7. The casting process is subdivided into two distinct subgroups: expendable and non-expendable mould casting.
8. Expendable mould casting is a generic classification that includes sand, plastic, shell, and investment (lost-wax technique) mouldings.

9. Non-expendable mould casting includes permanent, die, centrifugal, and continuous casting.
10. The casting process is used to produce a wide variety of parts, ranging from small components to entire body panels of cars.

Introduction to Patterns, Pattern Allowances, Ingredients of Moulding Sand and Melting Furnaces. Foundry Tools and Their Purposes Demo of Mould Preparation and Aluminum Casting Practice – Study and Preparation of Mould for Plastic

1. **Patterns**: A pattern is a model or replica of the object to be cast. It is used to create the cavity in the mould where the molten metal will be poured to form the casting.
2. **Pattern Allowances**: Pattern allowances are small adjustments made to the pattern to compensate for changes that occur during the casting process. These allowances include shrinkage allowance, draft allowance, machining allowance, and distortion allowance.
3. **Ingredients of Moulding Sand**: Moulding sand is a mixture of sand, clay, and water. The sand provides the structure for the mould, the clay binds the sand together, and the water helps to make the sand more workable.
4. **Melting Furnaces**: Melting furnaces are used to melt the metal that will be used in the casting process. There are several types of melting furnaces, including cupola furnaces, electric arc furnaces, and induction furnaces.
5. **Foundry Tools**: Foundry tools are used to prepare the mould and handle the molten metal. These tools include shovels, rammers, sprue cutters, and tongs.
6. **Demo of Mould Preparation and Aluminum Casting**: In this demo, a mould will be prepared using a pattern and moulding sand. The molten aluminum will then be poured into the mould to create a casting.
7. **Practice – Study and Preparation of Mould for Plastic**: In this practice, students will study the process of preparing a mould for plastic casting and then prepare a mould themselves.

7 CNC Shop 6 Hrs

1. CNC stands for Computer Numerical Control.
2. A CNC shop is a facility that uses CNC machines to manufacture parts and products.
3. CNC machines are controlled by computer programs, which allows for precise and accurate manufacturing.
4. A 6-hour shift in a CNC shop would involve operating and monitoring the CNC machines, loading and unloading materials, and performing quality checks on the manufactured parts.
5. CNC shops can produce a wide variety of parts and products, ranging from simple to complex designs.

6. CNC technology has revolutionized the manufacturing industry, allowing for faster and more efficient production.
7. Working in a CNC shop requires technical knowledge and skills, as well as attention to detail and the ability to follow instructions and procedures.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

CNC (Computer Numerical Control) machines are automated machines that are capable of performing a wide range of operations on metal components. These machines are controlled by computer programs and can be used to perform complex and precise operations with high accuracy and repeatability.

Some of the main features of CNC machines include:

1. **High precision and accuracy:** CNC machines are capable of performing operations with high precision and accuracy, thanks to their computer-controlled nature.

2. **Flexibility:** CNC machines can be programmed to perform a wide range of operations, making them highly flexible and versatile.

3. **Automation:** CNC machines are fully automated, which means that they can operate without the need for human intervention.

4. **High productivity:** CNC machines can operate at high speeds and can perform multiple operations simultaneously, resulting in high productivity.

Some of the main working parts of a CNC machine include:

1. **Control unit:** The control unit is the brain of the CNC machine. It is responsible for interpreting the instructions from the computer program and controlling the movements of the machine.

2. **Drive system:** The drive system is responsible for moving the various parts of the CNC machine, such as the spindle and the cutting tool.

3. **Spindle:** The spindle is the part of the CNC machine that holds and rotates the cutting tool.

4. **Cutting tool:** The cutting tool is the part of the CNC machine that performs the actual cutting operation on the metal component.

There are various accessories that can be used with CNC machines to perform different operations on metal components. Some of these accessories include:

1. **Tool holders:** Tool holders are used to hold the cutting tools in place on the spindle.

2. **Workholding devices:** Workholding devices are used to hold the metal component in place while it is being machined.

3. **Coolant systems:** Coolant systems are used to cool the cutting tool and the metal component during machining, to prevent overheating and damage.

CNC machines can be used to perform a wide range of operations on metal components, including drilling, milling, turning, and grinding. These operations can be performed with high precision and accuracy, resulting in high-quality metal components.

8 To prepare a product using 3D printing 3 Hrs

1. **Design the 3D model:** The first step in preparing a product using 3D printing is to design a 3D model of the product using computer-aided design (CAD) software. This model should be accurate and detailed, as it will serve as the blueprint for the final product.
2. **Choose the right 3D printing technology:** There are several different 3D printing technologies available, each with its own advantages and disadvantages. The most common technologies are Fused Deposition Modeling (FDM), Stereolithography (SLA), and Selective Laser Sintering (SLS). The choice of technology will depend on the desired properties of the final product, such as strength, flexibility, and surface finish.
3. **Prepare the 3D printer:** Before starting the printing process, the 3D printer must be prepared. This includes loading the appropriate material, calibrating the printer, and setting the printing parameters.
4. **Print the product:** Once the 3D printer is prepared, the printing process can begin. The 3D model is sliced into thin layers, and the printer builds the product layer by layer. The printing time will vary depending on the size and complexity of the product.
5. **Post-processing:** After the printing is complete, the product may require some post-processing. This can include removing support structures, sanding, and polishing to improve the surface finish.
6. **Quality control:** The final step is to inspect the product to ensure that it meets the desired specifications. This can include checking the dimensions, strength, and surface finish.

In summary, preparing a product using 3D printing involves designing a 3D model, choosing the right 3D printing technology, preparing the 3D printer, printing the product, post-processing, and quality control. The entire process can take around 3 hours, depending on the size and complexity of the product.

Course Outcome:

A course outcome is a statement that describes the knowledge, skills, and abilities that students are expected to have acquired by the end of a course. These outcomes are used to assess the effectiveness of the course and to ensure that students are meeting the learning objectives set by the instructor.

Some key points to consider when writing course outcomes include:

1. Outcomes should be specific, measurable, and achievable.
2. Outcomes should be aligned with the course content and objectives.
3. Outcomes should be written in clear and concise language.
4. Outcomes should be focused on what the student will be able to do or demonstrate by the end of the course.

By clearly defining course outcomes, instructors can better design course content and assessments to ensure that students are meeting the desired learning objectives. Additionally, course outcomes can help students understand what is expected of them and how they will be evaluated.

The students will be able to:

CO1. Use various engineering materials, tools, machines and measuring equipments. (Blooms Level: K3) - Students will learn to identify and select appropriate engineering materials for specific applications. - Students will become familiar with various tools, machines, and measuring equipment used in engineering. - Students will develop the ability to use these tools, machines, and measuring equipment effectively and safely.

CO2. Perform machine operations in lathe and CNC machine. (Blooms Level: K3) - Students will learn the principles and techniques of operating a lathe machine. - Students will become familiar with the operation of a CNC machine and its applications in manufacturing. - Students will develop the ability to perform basic machine operations on a lathe and CNC machine.

CO3. Perform manufacturing operations on components in fitting and carpentry shop. (Blooms Level: K3) - Students will learn the principles and techniques of fitting and carpentry. - Students will become familiar with the tools and equipment used in fitting and carpentry. - Students will develop the ability to perform basic manufacturing operations on components in a fitting and carpentry shop.

CO4. Perform operations in welding, moulding, casting and gas cutting. (Blooms Level: K3) - Students will learn the principles and techniques of welding, moulding, casting, and gas cutting. - Students will become familiar with the tools and equipment used in these operations. - Students will develop the ability to perform basic operations in welding, moulding, casting, and gas cutting.

CO5. Fabricate a job by 3D printing manufacturing technique. (Blooms Level: K3) - Students will learn the principles and techniques of 3D printing. - Students will become familiar with the tools and equipment used in 3D printing. - Students will develop the ability to fabricate a job using 3D printing manufacturing technique.

Reference Books:

Reference books are a type of non-fiction book that provides information on a specific subject. They are often used for research and study purposes and are designed to be consulted rather than read from cover to cover. Some common types of reference books include:

1. **Encyclopedias:** These books provide comprehensive information on a wide range of subjects, often in an alphabetical format. They are useful for obtaining general knowledge on a topic.
2. **Dictionaries:** These books provide definitions and meanings of words, often in alphabetical order. They are useful for understanding the meaning of unfamiliar words.
3. **Thesauri:** These books provide synonyms and antonyms for words, often in alphabetical order. They are useful for finding alternative words to use in writing.
4. **Atlases:** These books contain maps and geographical information. They are useful for understanding the location and geography of different places.
5. **Almanacs:** These books contain information on a wide range of subjects, often presented in a calendar format. They are useful for obtaining current information on various topics.
6. **Handbooks and Manuals:** These books provide detailed information and instructions on a specific subject. They are useful for learning how to do something or for understanding the details of a particular topic.

Reference books are often found in libraries and can be a valuable resource for students and researchers. They provide reliable and accurate information on a wide range of subjects and can be a useful tool for learning and understanding new information.

Workshop Practice by H S Bawa, published by McGraw Hill

Workshop Practice is a book written by H S Bawa and published by McGraw Hill. It is a comprehensive guide to the various aspects of workshop practice, including:

1. Introduction to workshop practice and safety measures.
2. Hand tools and their uses.
3. Measuring instruments and their applications.
4. Machine tools and their operations.
5. Welding and joining processes.
6. Sheet metal work.
7. Carpentry and pattern making.
8. Foundry and casting processes.
9. Fitting and assembly techniques.

This book is a valuable resource for students and professionals in the field of engineering and manufacturing, providing practical knowledge and skills for workshop practice. It is written in a clear and concise manner, with numerous illustrations and examples to aid understanding. It is a recommended text for courses in workshop practice and related subjects.

2. Mechanical Workshop Practice, K C John, PHI

Mechanical Workshop Practice is a book written by K C John and published by PHI. It covers the fundamental concepts and techniques used in a mechanical workshop. Some of the key topics covered in the book include:

1. Introduction to workshop practice
2. Safety precautions and first aid
3. Hand tools and their uses
4. Measuring instruments and their uses
5. Marking and layout tools
6. Cutting tools and their uses
7. Machine tools and their operations
8. Joining processes
9. Heat treatment processes
10. Surface finishing processes

This book is a valuable resource for students and professionals who want to learn about the practical aspects of mechanical engineering. It provides a comprehensive overview of the tools, techniques, and processes used in a mechanical workshop, and is written in a clear and concise manner that makes it easy to understand and follow.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice Vol 1 and Vol 2 are books written by S.K. Hajra Choudhury and published by Media Promoters of Publishers .
- These books cover the basics of workshop practices and manufacturing processes .
- The books are intended to help readers gain knowledge of the development of sheet metal models and their applications, as well as perform soldering and welding of different sheet metal and welded joints .
- The books are widely used as study material for exams in the field of mechanical engineering .

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

CNC stands for Computer Numerical Control. It is a technology that uses computer-controlled machines to perform various manufacturing operations.

CNC machines can be used for a wide range of applications, including milling, turning, drilling, and grinding.

Some key points to consider when studying CNC Fundamentals and Programming include:

1. Understanding the basic components of a CNC machine, including the control unit, machine tool, and workpiece.
2. Learning about the different types of CNC machines, such as milling machines, lathes, and grinders.
3. Understanding the role of programming in CNC machining, including the use of G-code and M-code.
4. Familiarizing oneself with the various tools and accessories used in CNC machining, such as cutting tools, tool holders, and workholding devices.
5. Learning about the different operations that can be performed on a CNC machine, including facing, contouring, and pocketing.
6. Understanding the importance of proper maintenance and calibration of CNC machines to ensure their accuracy and reliability.

This book, "CNC Fundamentals and Programming" by P. M. Agrawal and V. J. Patel, published by Charotar Publication, provides a comprehensive introduction to the subject and is a valuable resource for anyone looking to learn more about CNC technology.